Week 11: Polyphase Structures

Homework, Weekly Quiz, Complete Solutions, and Instructor Grading Notes

Course context: DSP + Machine Learning for a mixed senior / first-year ECE cohort. DSP reference: Oppenheim & Schafer §4.7–4.8. ML bridge: structured convolution and fixed/learned subband front ends.

Important precision point: The Noble identities do not allow an arbitrary filter to move through a rate changer. The downsampling identity applies to $G(z^M)$, and the interpolation identity applies to $G(z^L)$.

Learning targets

- Decompose an FIR filter into polyphase components and reconstruct the original impulse response.
- Explain the arithmetic savings of polyphase decimation and interpolation.
- Apply the First and Second Noble identities with the correct z^M / z^L conditions.
- Describe a two-channel analysis/synthesis filter bank and perfect reconstruction.
- Compare polyphase time-phase partitioning with grouped/depthwise CNN computation and fixed subband feature extraction.

Homework — Student Version

Problem 1 — Polyphase decomposition (12 pts) [Core]

Let $h[n] = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $M = 3$.
(a) Find $e_0[r]$, $e_1[r]$, and $e_2[r]$ using $e_k[r] = h[rM+k]$.
(b) Write $H(z)$ in polyphase form $H(z) = \Sigma_k z^{-k} E_k(z^M)$.
(c) Explain how the original tap ordering is recovered when the phases are recombined.

Problem 2 — Where the decimation savings comes from (10 pts) [Core]

A 48-tap FIR anti-aliasing filter precedes a downsampler by $M = 6$. The input rate is 96 kHz.
(a) Estimate the multiply rate for the naive filter-then-downsample implementation.
(b) Estimate the multiply rate for an ideal polyphase implementation.
(c) State the approximate speedup and explain why the number of multiplies per retained output is not reduced by the same factor.

Problem 3 — First Noble Identity and its limitation (12 pts) [Core + concept trap]

Let $G(z) = 1 + 2z^{-1}$ and $M = 2$. Use $x[n] = \delta[n]$.
(a) Compute the output of $G(z^2)$ followed by $\downarrow 2$.
(b) Compute the output of $\downarrow 2$ followed by $G(z)$. Verify equivalence.
(c) Now replace $G(z^2)$ in the left system by the arbitrary filter $H(z)=1+2z^{-1}$. Show that $H(z)\rightarrow\downarrow 2$ is not equivalent to $\downarrow 2\rightarrow H(z)$. Explain the lesson.

Problem 4 — Efficient decimator derivation (14 pts) [Senior derivation]

For y[m] obtained by filtering x[n] with h[n] and then downsampling by M, start from y[m]=Σ_q h[q]x[mM−q]. Write q=rM+k and derive a polyphase expression involving $e_k[r]=h[rM+k]$. What does the result say about which input sample phases are needed by each branch?

Problem 5 — Efficient interpolation phases (12 pts) [Core]

Let h[n] = {1, 2, 3, 4, 5, 6, 7, 8} and interpolation factor L = 4.

(a) Find the four polyphase impulse responses.

(b) Which phase produces output samples y[4m+2]?

(c) If the original FIR has N = 40 taps, compare the approximate multiply rates of naive zero-insertion + full-rate filtering and polyphase interpolation.

Problem 6 — Haar analysis/synthesis bank (14 pts) [Core / filter bank]

Use the pairwise Haar transform

a[m]=(x[2m]+x[2m+1])/√2 and d[m]=(x[2m]−x[2m+1])/√2.

For x = {3, 1, 2, −2}:

(a) Compute a[m] and d[m].

(b) Reconstruct x exactly using the synthesis equations.

(c) Verify that input energy equals total subband energy.

Problem 7 — Perfect-reconstruction reasoning (12 pts) [Senior concept]

A two-channel critically sampled analysis/synthesis bank uses analysis filters $H_0(z)$, $H_1(z)$ and synthesis filters $F_0(z)$, $F_1(z)$.

(a) In words, what two things must be true for perfect reconstruction?

(b) Interpret the alias-cancellation condition $F_0(z)H_0(-z)+F_1(z)H_1(-z)=0$.

(c) Why is “the two filters cover low and high frequencies” not by itself sufficient to guarantee perfect reconstruction?

Problem 8 — DSP/ML architecture synthesis (14 pts) [DSP + ML synthesis]

You must classify short vibration signals on a low-power embedded processor. Compare two front ends:

A. Four fixed FIR subbands followed by four log-energy features and a small classifier.

B. A learned Conv1d front end using grouped/depthwise convolution followed by a pointwise layer.

For each option discuss computation, trainable parameters, interpretability, adaptability, and one failure mode. Then recommend a design for a small labeled dataset and justify your choice. Finally, explain precisely how grouped convolution is similar to—but not mathematically the same as—polyphase decomposition.

Homework — Complete Solutions

Solution 1 — Polyphase decomposition (12 pts) [Core]

Solution

$$e_0 = \{1, 4, 7, 10\}, \quad e_1 = \{2, 5, 8\}, \quad e_2 = \{3, 6, 9\}$$

$$E_0(z) = 1 + 4z^{-1} + 7z^{-2} + 10z^{-3}$$

$$E_1(z) = 2 + 5z^{-1} + 8z^{-2}, \quad E_2(z) = 3 + 6z^{-1} + 9z^{-2}$$

$$H(z) = E_0(z^3) + z^{-1}E_1(z^3) + z^{-2}E_2(z^3)$$

The factor z^{-k} restores the phase offset k. Expanding $E_k(z^3)$ places its coefficients every third delay, and the outer z^{-k} shifts them back to indices k, k+3, k+6, … .

Grading guidance:

- 4 pts phase sequences; 5 pts z-domain form; 3 pts interpretation.

Solution 2 — Decimation savings (10 pts) [Core]

Solution

$$\text{naive} \approx 48 \cdot 96{,}000 = 4.608 \times 10^6 \text{ multiplies/s}$$

$$\text{polyphase} \approx (48/6) \cdot 96{,}000 = 0.768 \times 10^6 \text{ multiplies/s}$$

The ideal speedup is about 6×. Each retained output still depends on roughly all 48 FIR taps; the savings comes from computing only 1/6 as many output instants, not from using only 1/6 of the taps in a retained convolution sum.

Grading guidance:

- 3 pts naive rate; 3 pts polyphase rate; 4 pts explanation.

Solution 3 — First Noble Identity (12 pts) [Core]

Solution

(a) $G(z^2) = 1 + 2z^{-2}$, so with δ[n] the filter output is {1,0,2}; ↓2 retains {1,2}.

(b) δ[n] downsampled by 2 is still δ[m]. Filtering by $G(z) = 1 + 2z^{-1}$ gives {1,2}. The systems match.

(c) For $H(z) = 1 + 2z^{-1}$, H(z)δ={1,2}; ↓2 retains only {1}. But ↓2δ=δ followed by H gives {1,2}. They do not match. The Noble Identity requires the special $G(z^M)$ spacing; an arbitrary filter cannot simply be moved through ↓M.

Grading guidance:

- 3 pts part a; 3 pts part b; 4 pts counterexample; 2 pts lesson.

Solution 4 — Efficient decimator derivation (14 pts) [Core]

Solution

$$y[m] = \Sigma_q\, h[q]\, x[mM-q]$$
$$q=rM+k \Rightarrow y[m]=\Sigma_{k=0}{}^{M-1}\, \Sigma_r\, h[rM+k]\, x[mM-rM-k]$$
$$y[m]=\Sigma_k\, \Sigma_r\, e_k[r]\, x[(m-r)M-k]$$

For branch k, the input samples have indices congruent to −k modulo M (with the exact apparent phase depending on the chosen delay/commutator drawing). Each E_k filter can therefore operate on a phase-selected low-rate sequence rather than on the full-rate stream. Summing the branch outputs reproduces exactly the retained full-rate convolution samples.

Grading guidance:

- 5 pts index substitution; 5 pts final expression; 4 pts phase/rate interpretation.

Solution 5 — Efficient interpolation phases (12 pts) [Core]

Solution

$$e_0=\{1,5\},\quad e_1=\{2,6\},\quad e_2=\{3,7\},\quad e_3=\{4,8\}$$

Phase e_2 produces the outputs y[4m+2]. In general, $y[mL+k]=(e_k*x)[m]$ for the usual causal phase ordering.

For N=40 and L=4, naive full-rate FIR work is approximately $40\cdot4\cdot F_s = 160F_s$ multiplies/s. Polyphase work is approximately $40F_s$ multiplies/s total, an ideal 4× saving.

Grading guidance:

- 4 pts phases; 3 pts output phase; 5 pts operation-rate comparison.

Solution 6 — Haar filter bank (14 pts) [Core]

Solution

$$a[0]=(3+1)/\sqrt{2} = 2\sqrt{2},\quad d[0]=(3-1)/\sqrt{2} = \sqrt{2}$$
$$a[1]=(2-2)/\sqrt{2} = 0,\quad d[1]=(2-(-2))/\sqrt{2} = 2\sqrt{2}$$

Reconstruction: $x[0]=(2\sqrt{2}+\sqrt{2})/\sqrt{2}=3$, $x[1]=(2\sqrt{2}-\sqrt{2})/\sqrt{2}=1$, $x[2]=(0+2\sqrt{2})/\sqrt{2}=2$, $x[3]=(0-2\sqrt{2})/\sqrt{2}=-2$.

$$\text{input energy} = 3^2+1^2+2^2+(-2)^2 = 18$$
$$\text{subband energy} = (2\sqrt{2})^2+(\sqrt{2})^2+0^2+(2\sqrt{2})^2 = 8+2+0+8 = 18$$

The orthonormal Haar transform is energy preserving and perfectly reconstructing.

Grading guidance:

- 5 pts analysis coefficients; 5 pts reconstruction; 4 pts energy check.

Solution 7 — Perfect reconstruction (12 pts) [Core]

Solution

- Aliasing created by the downsamplers must cancel when the synthesis branches are added.
- The desired non-aliased transfer path must reduce to only a constant scale and pure delay: $\hat{x}[n]=c\cdot x[n-d]$.
- The condition $F_0(z)H_0(-z)+F_1(z)H_1(-z)=0$ says that the folded/aliased contribution from one branch is exactly canceled by the folded contribution from the other.
- Frequency coverage alone is insufficient because the filters also have phase, transition-band, amplitude, and synthesis relationships. Two filters can split low/high energy yet still leave aliasing or amplitude/phase distortion after reconstruction.

Grading guidance:

- 4 pts two requirements; 4 pts alias-condition interpretation; 4 pts why coverage is insufficient.

Solution 8 — DSP/ML architecture synthesis (14 pts) [Core]

Solution

- Fixed four-band front end: very few or no trainable front-end parameters, predictable computation, interpretable band energies, and good data efficiency. Failure mode: the chosen bands may discard task-relevant structure or be mismatched to a new operating condition.
- Grouped/depthwise Conv1d: fewer parameters/MACs than dense convolution and can adapt filters to the task; a pointwise layer restores cross-channel mixing. Failure modes include learned aliasing when stride is used, overfitting with limited labels, or learned filters that are difficult to interpret.
- For a small labeled dataset on a tight embedded budget, a fixed DSP bank plus tiny classifier is a strong baseline. A hybrid—DSP initialization followed by limited fine-tuning—can be tested if enough validation data exists.
- Similarity: both exploit structured independent branches and later recombine results. Difference: polyphase partitions taps/samples by time phase modulo the rate-change factor and is an exact reorganization of a multirate FIR. Grouped convolution partitions feature channels and is a learnable architectural constraint; it is not a Noble-identity/polyphase transformation.

Grading guidance:

- 4 pts fixed front end; 4 pts grouped/learned front end; 3 pts recommendation; 3 pts precise analogy/difference.

Weekly Quiz — Student Version

Suggested time: 15–20 minutes. Total: 20 points. Basic calculator allowed.

1. (2 pts) For M=3, write the definition of the k-th polyphase sequence $e_k[r]$ in terms of h[n].

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2. (2 pts) A 30-tap FIR is followed by ↓5. Approximately what ideal arithmetic speedup does a polyphase decimator provide relative to filtering every full-rate output?

_____

3. (3 pts) True or false: H(z)→↓M is always equivalent to ↓M→H(z). Explain.

_____

4. (3 pts) For h={1,2,3,4,5,6} and M=2, find e_0 and e_1.

_____

5. (2 pts) State the Second Noble Identity in words or block-diagram form.

_____

6. (3 pts) A 24-tap FIR interpolates by L=3. Compare the approximate naive and polyphase multiply rates in units of F_s.

_____

7. (2 pts) In a two-channel perfect-reconstruction bank, what two broad requirements must be satisfied?

_____

8. (3 pts) Give one similarity and one important difference between polyphase decomposition and grouped convolution.

_____

Weekly Quiz — Solutions and Grading

1. Polyphase definition

$e_k[r]=h[rM+k]$, $k=0,\ldots,M-1$.

Grading: 2 pts.

2. Speedup

Approximately 5×.

Grading: 2 pts; accept explanation that only one of five output instants survives.

3. Noble-identity trap

False. A filter may commute across ↓M in the Noble identity when it has the special $G(z^M)$ form before the downsampler; arbitrary $H(z)$ does not commute.

Grading: 1 pt false; 2 pts condition/explanation.

4. M=2 decomposition

$e_0=\{1,3,5\}$, $e_1=\{2,4,6\}$.

Grading: 1.5 pts each.

5. Second Noble Identity

↑L followed by $G(z^L)$ is equivalent to $G(z)$ followed by ↑L.

Grading: 2 pts.

6. Interpolation work

Naive ≈ $24\cdot 3F_s=72F_s$ multiplies/s; polyphase ≈$24F_s$; about 3× saving.

Grading: 1 pt each rate; 1 pt speedup.

7. Perfect reconstruction

Alias terms cancel, and the desired transfer path is only a scale and delay (no frequency-dependent distortion).

Grading: 1 pt each.

8. Polyphase vs grouped conv

Similarity: both partition work into parallel structured branches. Difference: polyphase partitions time/tap phases and exactly reorganizes a multirate FIR; grouped convolution partitions learned feature channels.

Grading: 1.5 pts similarity; 1.5 pts difference.

Instructor Notes and Differentiation

For first-year students

- Use tap-index coloring and explicit M=2 or M=3 sequences before z-domain notation.
- Grade the Noble identities primarily as “special spaced filter can move to the low-rate side,” rather than requiring a formal proof.
- Use the Haar pairwise average/difference example as the main filter-bank anchor.

For senior students

- Require the q=rM+k decimator derivation and careful input-phase indexing.
- Require the alias-cancellation / distortion-transfer equations for two-channel perfect reconstruction.
- Extension: derive a polyphase matrix representation for the analysis bank or implement a custom decimator and compare timing against scipy.signal.upfirdn.